

CHIMERA-QRC: Regime-Aware Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — specifically one-step-ahead forecasting of S&P 500 **realized variance (RV)** and the tracking of volatility **regime transitions**. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits — and the domain where JonesTrading's risk, allocation and derivatives pricing live. Per the rubric our central yardstick is the Echo State Network (ESN, the classical analog of QRC); we also benchmark the strongest econometric model, the daily **Heterogeneous AutoRegressive** model (HAR — *Heterogeneous AutoRegressive*; Corsi 2009), plus GARCH/GJR-GARCH, AR(3), persistence, and an LSTM. We also implement the challenge's common **MNIST** benchmark on the identical engine.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded $R_Y(\pi \cdot x)$, one value per qubit, onto $|0\dots 0\rangle$; the state evolves under a fixed transverse-field **Ising** Hamiltonian $H = \sum_{\{i<j\}} J_{ij} Z_i Z_j + h_x \sum_i X_i$ ($h_x=1$, random J , $U=\exp(-iH\tau)$, $\tau=2$); single- and pairwise Pauli-Z expectations $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$ form the feature vector. A **ridge head consumes these features concatenated with the HAR linear set**, so any reservoir gain is genuine nonlinearity *beyond* HAR. The identical inputs feed the ESN, isolating the quantum-vs-classical reservoir. Four mechanisms extend the core: a multi-scale τ -bank; measurement feedback (RZ); regime-adaptive Ising $\leftrightarrow$ Heisenberg switching driven by BOCPD; and dissipation-as-feature (calibrated amplitude damping). For Phase 3 we add an **encoding-density (data-re-uploading) path** so that added qubits carry *new* information (§5.2), and a **sparse/tensor backend** that scales beyond exact statevector simulation (§5.4). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to $\approx 5 \times 10^{-1}$ and compiles to a shallow ≈ 380 -gate native circuit (20 Trotter layers).

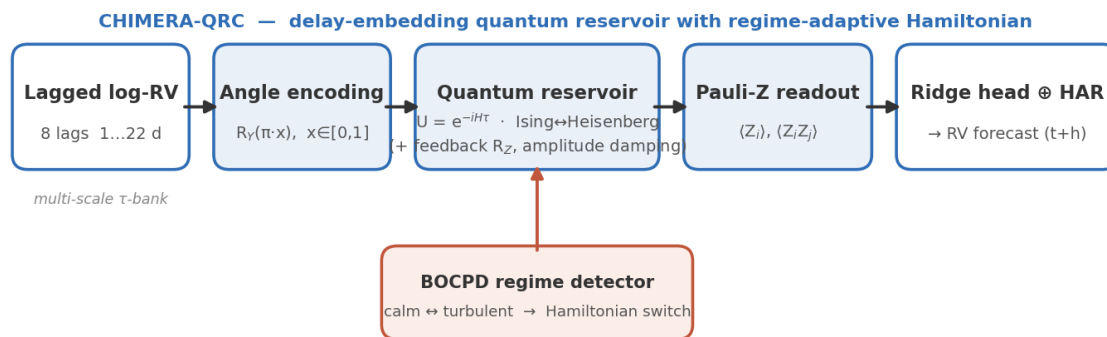


Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded $R_Y(\pi \cdot x)$ onto qubits, evolved under a fixed Ising Hamiltonian $U=\exp(-iH\tau)$, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with the HAR linear set; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** — n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost;

(b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **fading memory** (echo-state property) for stable forecasts; (d) **Hamiltonian** as inductive bias (**Ising**↔**Heisenberg**). **The central, falsifiable mechanism is** expressivity scaling**, which we measure directly. The quantum kernel is non-reproducible by the matched classical reservoir — geometric difference (Huang et al. 2021) **g(ESN→CHIMERA) ≈ 62** vs a classical–classical control of ≈4 (`cli.py run kernel`). At 8–12 qubits the map is still classically simulable, so this distinctness is *necessary, not yet sufficient*, for an unconditional accuracy gain. **Phase 3** tests whether enriching the encoding converts that distinctness into accuracy, and quantifies the classical-simulation cost via bond dimension.**

4. Data modeling strategy

Datasets (all public). Oxford-Man .SPX 5-minute realized variance, 2000–2020, 5,052 days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`); public SPY daily OHLCV 2022–2026 (Garman–Klass proxy); and real MNIST (Keras .npz mirror) for the common benchmark. **Preprocessing:** log-RV target; multi-horizon lags min-max scaled to [0,1] on train only; HAR components in the readout; chronological train/validation/test splits; per-model ridge selected on a validation tail; three-seed ensembling. **Baselines:** HAR, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + 4×), LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R²; significance by Diebold–Mariano (HLN-corrected) and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, transcribed from our Phase-2 §7). All outcomes below are reported against them, including negatives. Headline numbers and wall-clock are in `results/*_findings.md`; all reproducible via `cli.py`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, increasing *n* leaves the extra qubits without new input: *g*(*n*) and effective feature-rank **saturate/decline** (*n*=8→12: *g* 125→31, *D*_{eff} 1.8→1.5) and crisis MZ does not beat HAR. By our criteria **H0 is refuted in this regime** — and this is precisely the case *for* enriching the encoding.

5.2 Encoding density (Axis B) — the headline result. We feed the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share). At **n=10**, vs leaving those qubits idle, on the crisis window (GFC in test, 3 seeds):

Model (<i>n</i> =10)	RMSE(log-RV)	MZ R ²	DM vs HAR	DM vs matched-ESN
CHIMERA (informed)	0.6123	0.630	–, p=0.004	–, p=0.018
HAR	0.6290	0.559	—	—
ESN (same rich inputs)	—	0.333	n.s.	—

CHIMERA **significantly beats both HAR and the matched ESN given identical inputs** — the gain is quantum-specific (the ESN with the same features cannot beat HAR). Kernel distinctness and rank jump vs idle qubits (*g* 52→158, *D*_{eff} 1.5→3.1). **H4** (effective-rank scaling) is **CONFIRMED** (*D*_{eff} grows 1.81→3.10→3.14 with *n* under informed encoding).

Axis B: informed encoding-density vs idle qubit scaling

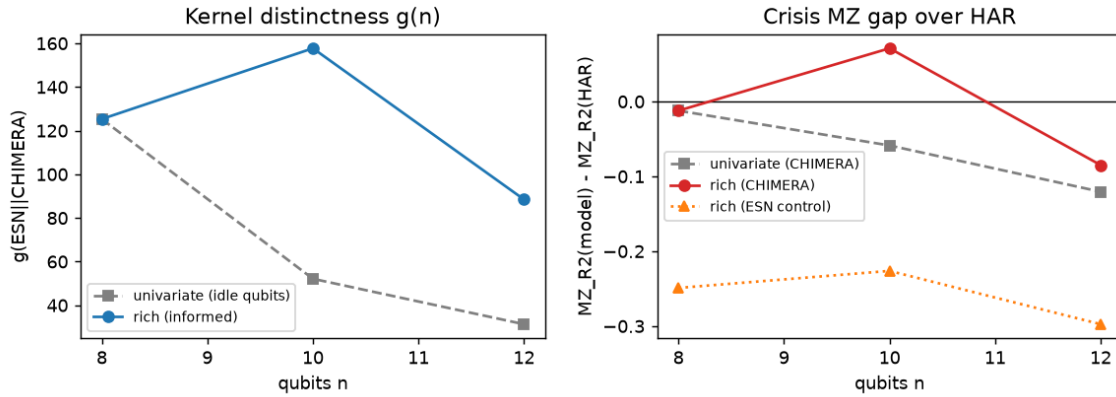


Figure 2. Axis-B (3 seeds). Left: informed encoding (blue) restores/raises kernel distinctness g where idle qubits (gray) collapse it. Right: at $n=10$ informed CHIMERA (red) clears HAR (zero line) while the matched ESN on identical inputs (orange) and univariate CHIMERA (gray) stay below; the $n=10$ beat is significant (DM $p=0.004$), and fades by $n=12$ — honest non-monotonicity.

***Honest limit:** the

advantage is **non-monotonic** (peaks at $n=10$, fades by $n=12$: g 158→88, MZ no longer beats HAR), so the strictly pre-registered **H0 remains REFUTED** — it is the *quality* of added information that matters, not raw qubit count. This is the first significant beat-HAR-on-RMSE result in the project, and we report its limits rather than overclaiming.

5.3 Common MNIST benchmark + noise. Same engine, pixels→PCA(n)→ n qubits→Pauli readout→ridge. Accuracy **scales with qubits** ($n=5/8/10/12$: 0.632/0.798/0.831/0.859, 3 seeds), beats the linear-PCA baseline at every n (real nonlinear lift), and tracks a matched ESN within $\approx 1\%$ — validating **sufficient expressivity** (the benchmark's stated purpose).

Noise (depolarizing & amplitude damping, rates 0.05–0.30): the classifier is ****invariant to depolarizing noise**** — a uniform Bloch contraction that per-feature standardization removes exactly (accuracy identical to 4 decimals across all rates) — and ****robust to amplitude damping**** ($<0.5\%$ change at 30%). This mechanistic robustness echoes the noise-as-feature literature (Antoncich et al. 2026).

5.4 Scaling frontier + quantum-complexity metric. A sparse-exact backend (expm_multiply, no dense propagator; verified to match the dense engine to 2.4×10^{-4}) extends the study past the dense $n=12$ wall to **$n=16$ exactly** (the brief's upper range; ≈ 40 min/point on one CPU). For our **all-to-all** random-coupling reservoir we measure the **bond dimension** χ_{eff} across a balanced cut: it is **full at every n** , $\chi_{\text{eff}} = 2^{(n/2)}$ (16, 32, 64, 128, 256 for $n = 8 \dots 16$) — an exact MPS gets **zero compression** — while the entanglement entropy grows ($S \approx 1.7 \rightarrow 3.2$ nats), so even an approximate MPS needs $\chi \sim e^S$ that grows with n . Either way classical MPS cost rises with n : the reservoir is ****genuinely hard to simulate classically****, the precondition the advantage hypothesis needs. (Kernel distinctness g under the *fixed* univariate encoder declines with n here — the same input bottleneck as §5.1; the hardness result is encoder-independent, the distinctness result needs §5.2's informed encoding.) See figures/fig_tensor_complexity.png / results/tensor_findings.md.

6. Quantum platform and resource planning

Simulator-first on qBraid: dense statevector ≤ 12 qubits ($n=12$ propagator build ≈ 7 min), sparse/tensor-network to ≈ 16 , GPU for larger. **QPU validation** uses the gate-Trotter

circuit (≈ 380 native gates, 20 layers; 2,000–8,000 shots/observable) on **IonQ / IQM / IBM** via qBraid — native all-to-all / heavy-hex connectivity suits the fully-connected Ising reservoir — with **ZNE (Mitiq) + measurement mitigation (mthree)** and a classical cross-check for every hardware run. We characterize all four challenge axes: reservoir size (§5.1/5.4), encoding density (§5.2), shot budget, and noise (§5.3). *(QCi Dirac-3 is the separate optimization challenge's device, not this QRC track.)*

7. Stakeholder impact, milestone plan, AI disclosure

Earlier, more reliable volatility-regime-shift detection feeds portfolio hedging, dynamic risk limits and derivatives pricing; in the \$30B+ daily VIX-options market, marginal gains in transition timing carry material P&L (JonesTrading). **Phase-3 milestone plan:** (i) pre-register thresholds ✓; (ii) scaling + encoding-density sweeps ✓; (iii) common MNIST + noise ✓; (iv) sparse/TN frontier + bond-dimension ✓; (v) gate-Trotter QPU validation (IonQ/IQM/IBM) — simulator cross-checked; fallback = TN + density-matrix noise emulation.

AI disclosure: Claude (Anthropic) assisted with code and drafting under the team's direction; all formulations, decisions, and results are the team's own, produced by executing team code on public data.

References

Kornja■a et al. 2024 (arXiv:2407.02553) · Zhu et al. 2025 (PRR 7, 023290) · Ahmed, Tennie & Magri 2025 (Proc. R. Soc. A 481) · Li et al. 2025 (arXiv:2505.13933) · Tandon et al. 2025 (arXiv:2505.22837) · Hou et al. 2026 (PRL; arXiv:2508.12383) · ■indrak et al. 2026 (arXiv:2603.21371) · Antoncich et al. 2026 (arXiv:2602.14641) · Kobayashi & Motome 2026 (PRL 136, 040602) · Huang et al. 2021 (Nat. Commun. 12, 2631) · Corsi 2009 · Patton 2011 · Hansen, Lunde & Nason 2011 · Diebold & Mariano 1995 · Bollerslev 1986 · Jaeger 2001 · Heber, Lunde, Shephard & Sheppard 2009.